

The Architecture of Affect

Threat, Opportunity, and Repair
Regulated by Three Neuropeptides of POMC

ACTH · Mobilization

β -endorphin · Conservation

α -MSH · Recalibration

Introduction to Meta-strophe Theory · Part One

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If affect is central, what architecture regulates it?

The congress claim becomes a substrate-free functional-architecture question: how affect coordinates regulation across brain and body.

Functional architecture

What structure makes affect regulatory across substrates?

Brain-body communication

How is organismic state broadcast, compared, and updated?

Coordinated response

How are regulatory actions selected, fueled, and revised?

Clinical implications

How does this change diagnosis, treatment, and prognosis?

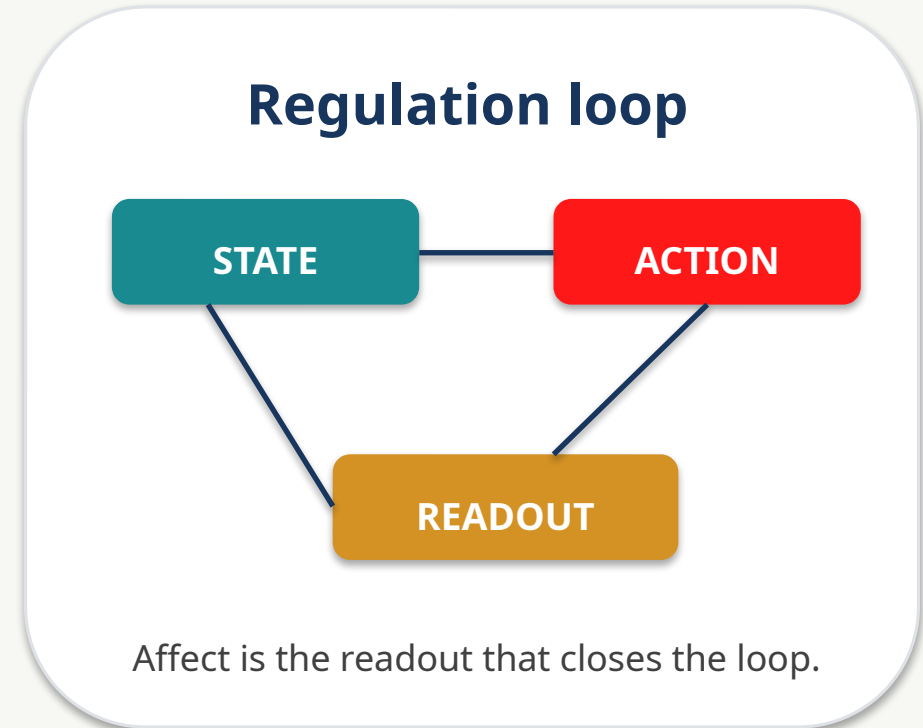
Solms and Friston place affect at the foundation of consciousness and press the question of functional organization. This talk proposes one biological candidate: POMC-derived regulatory address space.

Affect is the felt face of regulation

Not just a feeling added to physiology — a readout of regulatory action.

The organism has to register:

- 1 What state it is in**
- 2 What response that state can support**
- 3 Whether it has enough supply to meet the demand**



POMC is ancient, not cortical

A precursor molecule cleaved into regulatory peptides that coordinate brain and body.

POMC

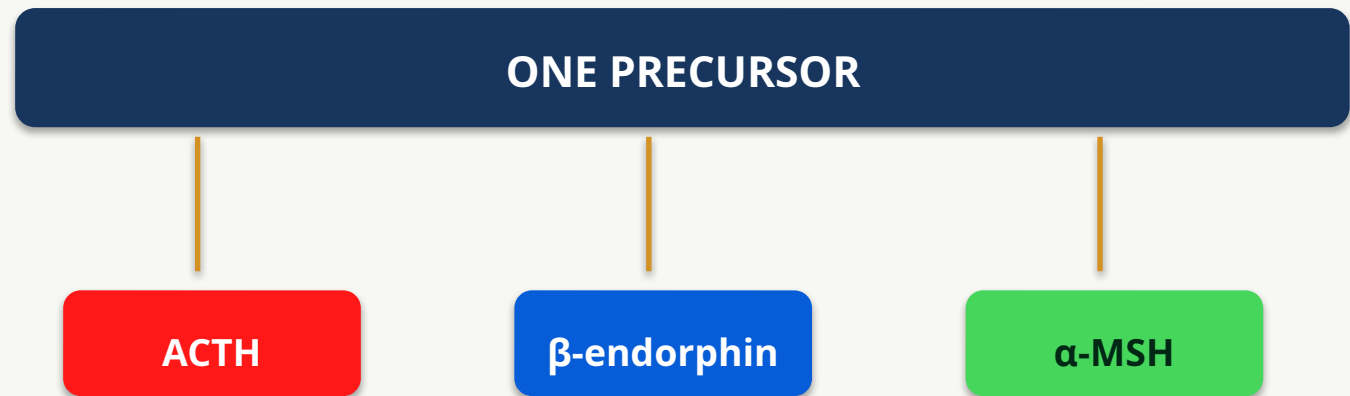
proopiomelanocortin

Conserved across vertebrate evolution

Cleaved on demand

Coordinates state across body and brain

A candidate molecular-affective grammar



Three regulatory economies. One condensed communication: affect state.

One precursor, three regulatory economies

The peptide triplet becomes the first visual grammar of the talk.

ACTH

Mobilization

- cortisol / glucose
- cardiovascular activation
- readiness for action

β -endorphin

Conservation

- analgesia / endurance
- protective withdrawal
- shielding when action is too costly

α -MSH

Recalibration / repair

- energy balance
- immune / inflammatory tuning
- safe strategy update

Maintain viability under constraint, at minimum cost

Before psychology or cognition, living systems face a thermoeconomic problem.



Three coordinates define a regulatory address

Affective state becomes legible as a point in structured regulatory space.



Valence

Threat or opportunity?
The frame in which
regulatory fuels are used.



Demand

Mobilize, conserve, or
recalibrate?
The strategy being
implemented.



Supply

Peptide-specific fuel
relative to demand.
Mismatch generates
affective state.

Notation: $V \pm [D_X \text{ relation } S_Y]$ **example:** $V - [D_M > S_ACTH]$

A regulatory address becomes a point in space

The same grammar can be read clinically, biologically, and geometrically.

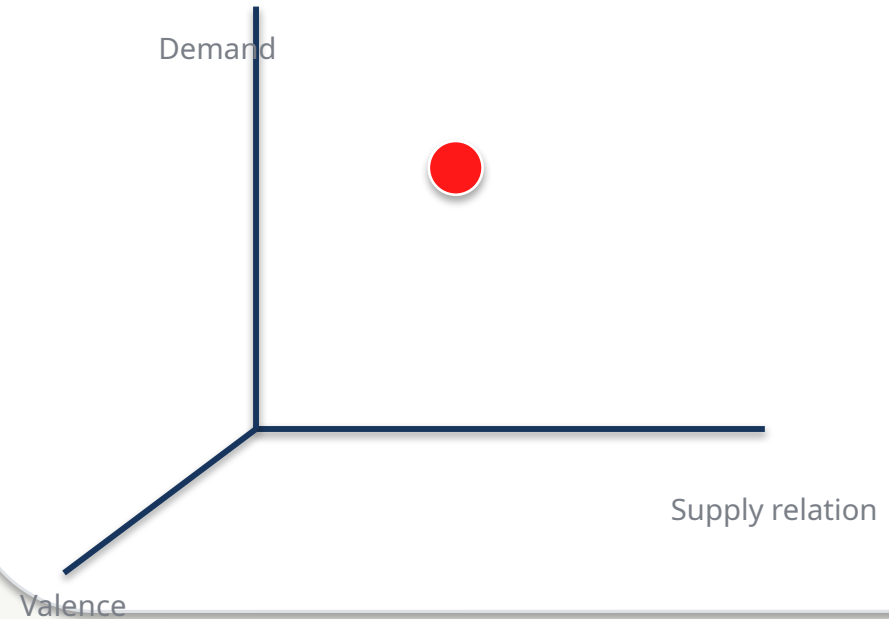
Regulatory address

V-[D_M > S_ACTH]

Threat frame
Mobilization demand
ACTH undersupply

ANXIETY

Point in regulatory space



Each named affect is a location, not merely a label.

The geometry that falls out is a sphere

Catastrophe is the turn downward; Meta-strophe is the turn beyond.



Part One: the structure of affect.
The larger theory asks how organisms turn beyond failed regulation.

First: a single axis — demand

Before affect labels, the system must choose a regulatory strategy.

MOBILIZE

do more · spend energy · act

**This choice applies
whether the situation
is later read as threat
or opportunity.**

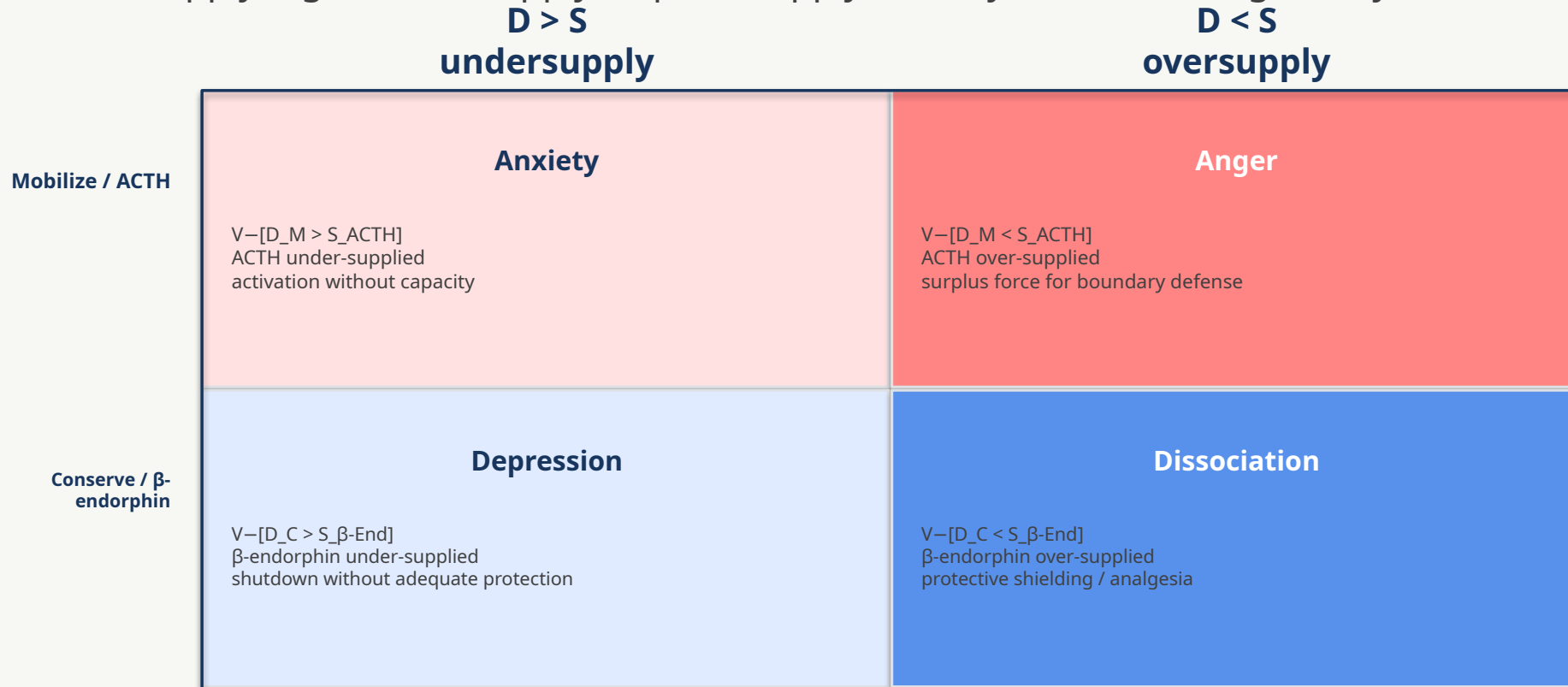
do less · reduce expenditure ·
shield

CONSERVE

**Valence comes
next.**

Threat compresses regulation into a 2×2 map

Left is undersupply; right is oversupply. Peptide supply is always relative to regulatory demand.



Threat = act now or shield now. No true recalibrative middle is foregrounded.

Threat × demand × supply produces clinical states

Same coordinate grammar; four familiar clinical configurations.

Anxiety

$V-[D_M > S_ACTH]$
ACTH under-supplied

Anger

$V-[D_M < S_ACTH]$
ACTH over-supplied

Depression

$V-[D_C > S_β-End]$
β-endorphin under-supplied

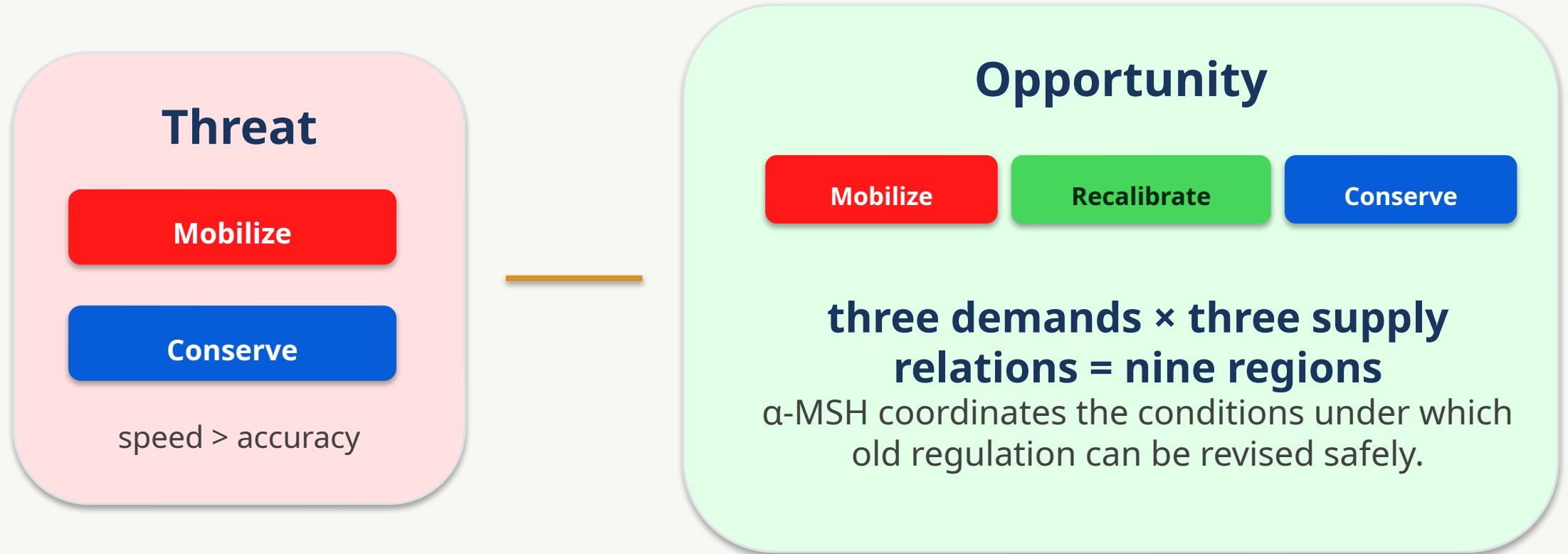
Dissociation

$V-[D_C < S_β-End]$
β-endorphin over-supplied

The map distinguishes activation without capacity, forceful defense, unprotected shutdown, and protective shielding.

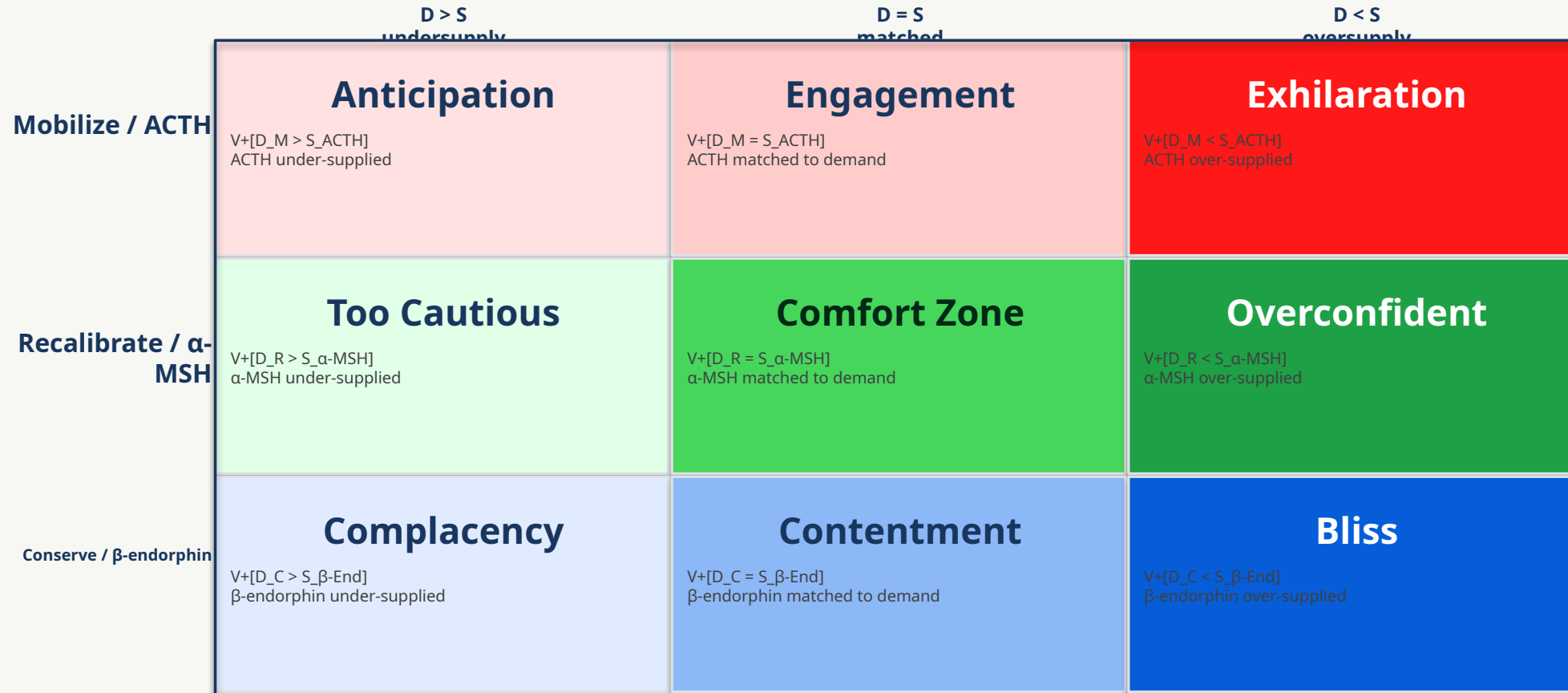
α -MSH adds the third regime: recalibration

Opportunity does not mean “pleasant”; it means enough room to update rather than merely react.



The opportunity face opens nine regions

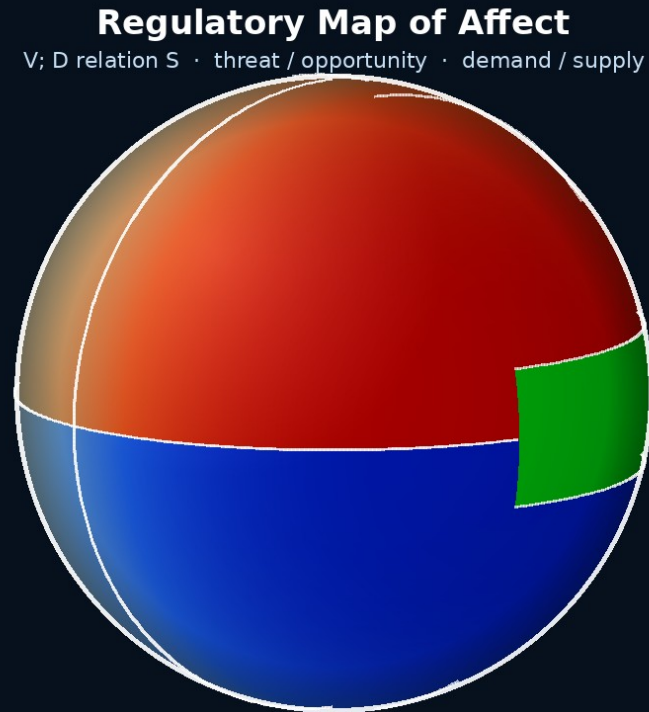
Opportunity adds matched supply states; recalibration creates the middle band.



Comfort Zone = maximum regulatory freedom: the system can mobilize, conserve, and update without collapsing back into threat.

Entering the regulatory map

If the live/local sphere works, this is where the deck gives way to the model.



Sphere sequence

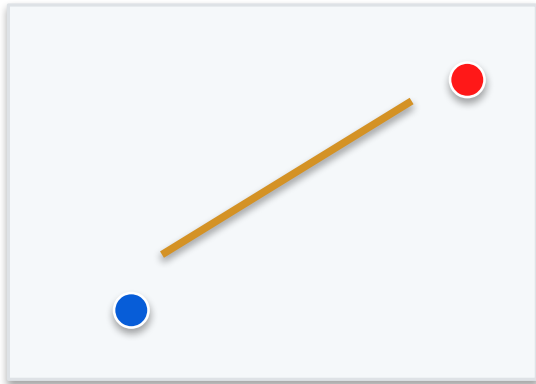
1. Threat face: demand + supply
2. Four threat affects
3. Opportunity face: α -MSH adds recalibration
4. Clinical trajectories

Backup slides follow if AV fails.

Diagnosis is a trajectory, not a box

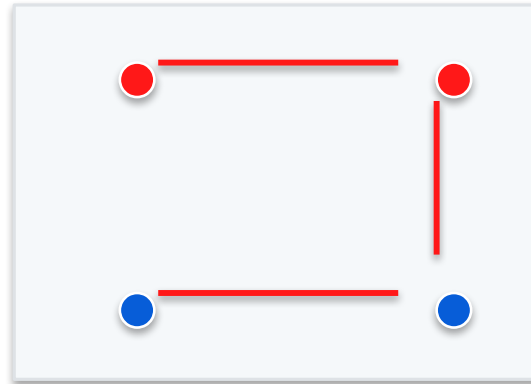
Repeated mismatches move the system through regulatory space.

Bipolar disorder



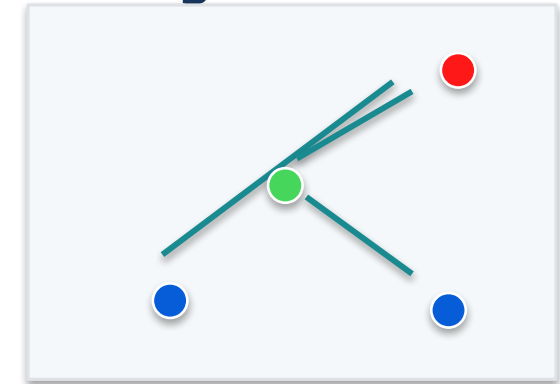
diagonal traversal: over-supplied mobilization → depleted conservation

PTSD



threat-lock: cycling within the threat face, recalibration unavailable

Borderline organization



rapid cross-valence instability, difficulty stabilizing in matched zone

Three convergences, one proposed architecture

Each mapping is a bridge, not the full paper.

Freud

- Pleasure principle: biphasic discharge
- Reality principle: triphasic recalibration
- Nirvana principle: zero-demand origin

Panksepp

- Primary emotional systems
- Motivational vectors
- Direction + intensity + resource signature

Polyvagal Theory

- Mobilization and shutdown as threat strategies
- Social engagement as regulated access to safety
- Meta-strophe adds peptide-specific supply and recalibration

The claim is structural convergence: prior models describe parts of a larger regulatory architecture.

Three claims to test under pressure

If the trajectories do not fit the cases, the model should change.

1

Affective states are regulatory addresses: valence, demand, and peptide-specific supply.

2

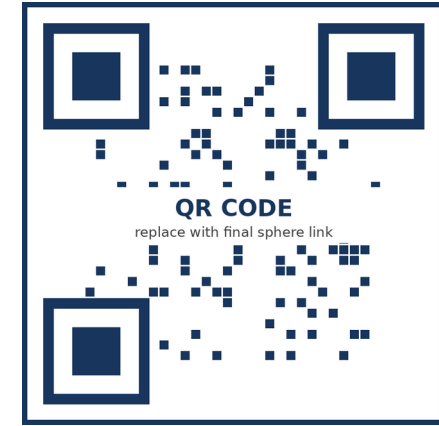
Clinical categories are trajectories through the sphere, not flat symptom clusters.

3

Treatment alters trajectories by restoring recalibration and reducing threat-lock.

Rotatable map · clickable regions · clinical trajectory overlays

Further developments and resource pages will be posted as completed.



**Enter the
sphere**